SEID 2364 – Assignment 8 Bias, Decision-Making, and Harm in AI-Mediated Systems

Part 1 – System Definition

Selected System: AI Hiring System

The AI hiring system is designed to process resumes, rank applicants, and assist recruiters during recruitment decisions. The system uses machine learning models to analyze applicant qualifications, work experience, educational background, and assessment results.

Inputs include resumes, applicant records, and evaluation criteria. Outputs include rankings, recommendations, or filtering decisions used by recruiters.

Part 2 – Bias and Fairness Synthesis

Bias in AI systems occurs when decisions unfairly favor or disadvantage certain individuals or groups. Fairness involves ensuring that people are treated equally and evaluated using reasonable and transparent standards.

One tension between implicit bias and algorithmic fairness is that human social bias can become embedded into datasets and later reproduced by algorithms. Even if developers attempt to create fair systems, historical data may already contain discrimination.

The ethics of search and ranking systems also helps explain risks in AI hiring. Ranking systems shape visibility and opportunities. If applicants are ranked unfairly, some individuals may lose opportunities before human review even occurs.

Part 3 – Deployment Scenarios

Scenario 1 – Corporate Recruitment

A large company uses the AI system to screen applicants for full-time jobs. The AI acts as an advisory tool for recruiters.

What changed: The system operates at a large scale with high consequences because employment decisions affect long-term income and career growth.

Scenario 2 – University Internship Selection

A university career office uses the system to rank internship applicants.

What changed: The user group changes from professional workers to students. The consequences involve educational and career opportunities rather than direct long-term employment.

Scenario 3 – Internal Employee Promotion Screening

A company uses the same system internally to recommend employees for promotion opportunities.

What changed: The deployment context changes because the system now evaluates existing employees rather than external applicants.

Part 4 – Mediation Pathway and Bias Location

Deployment 1 – Corporate Recruitment

Applicant Data → AI Screening → Recruiter Review → Hiring Outcome

Bias Risks: - Historical hiring bias in training data - Proxy variables linked to gender or ethnicity - Recruiter overreliance on AI recommendations

Deployment 2 – Internship Selection

Student Resume → AI Ranking → Career Office Review → Internship Recommendation

Bias Risks: - Bias toward prestigious universities - Limited work experience affecting rankings - Unequal access to extracurricular opportunities

Deployment 3 – Promotion Screening

Employee Records → AI Evaluation → Management Review → Promotion Decision

Bias Risks: - Bias from past performance evaluations - Reinforcement of workplace inequality - Reduced transparency in promotion decisions

Comparative Analysis

Some bias locations exist across all deployments, especially within training data and ranking models. However, deployment context changes the type of harm produced. Internship systems may disadvantage inexperienced students, while promotion systems may reinforce workplace inequality.

Part 5 – Decision Pathway Analysis

The AI system produces rankings or recommendations that influence decision-makers. Recruiters, managers, or career offices then use those outputs to make final decisions.

Human judgment enters when reviewers interpret AI recommendations. However, if humans trust the system too strongly, the AI may effectively control decisions indirectly.

If outputs are biased or incomplete, individuals may lose employment or career opportunities unfairly.

Part 6 – Harm Analysis

Direct harms include discrimination, economic loss, and unfair rejection of applicants.

Indirect harms include reduced trust in hiring systems and reinforcement of social inequality. If biased systems are widely adopted, discrimination may become normalized at a larger societal scale.

Different groups experience harm differently. Marginalized communities may face greater disadvantages because existing inequalities can become amplified through AI systems.

Conclusion

This assignment showed that bias can enter AI systems at many stages, including data collection, ranking processes, deployment context, and human interpretation.

The same AI system may produce different harms depending on how and where it is used. Ethical analysis therefore requires examining not only the technology itself but also the surrounding institutional and social environment.